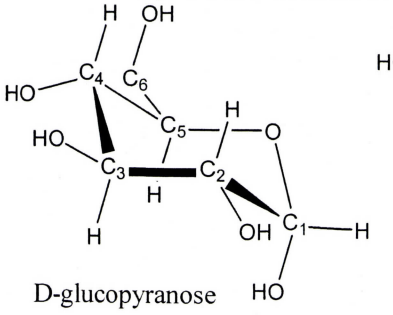
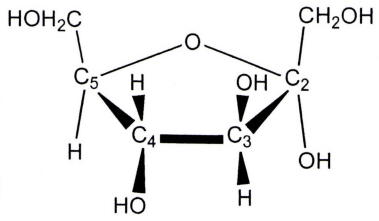
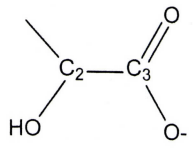
Biochemistry Cume 4/28/2017



D-glucopyranose



D-fructofuranose



lactate

1. (33 pts) Fructose is labeled with ^{14}C at C-2 is incubated with glycolytic enzymes to convert necessary cofactors. Hexokinase can also convert D-fructose to fructose 6-phosphate. Which carbon atom(s) of fructose 1,6-bisphosphate, dihydroxyacetone phosphate, glyceraldehydes 3-phosphate, 1,3-bisphosphoglycerate and pyruvate would be labeled? Give metabolic pathways for these reactions, and show your work.

2. (33 pts) Fructose labeled with ^{14}C at C-2 is added to a cell extract containing the enzymes and cofactors of the glycolytic pathway, the citric acid cycle, and the pyruvate dehydrogenase complex? Draw the correct structure and show labeled carbons in the following intermediate molecules: citrate, alpha-ketoglutarate, fumarate and oxaloacetate.

3. (34 pts) Given a mitochondrial extract with all of the enzymes and cofactors normally found inside mitochondria but **lacking pyruvate dehydrogenase completely,** show how pyruvate, CO_2, and acetyl CoA can be used to form succinate when succinate dehydrogenase is inhibited by malonate. Assume there is no oxaloacetate initially present. Draw the correct structure for each intermediate molecule. Include the names or abbreviations of any other reactants and products (eg, ATP, ADP, Pi, NAD+, NADH, etc). Show the path of ^{14}C label in each molecule above starting with ^{14}C label in the **C-2** of pyruvate (CH_3-C(O)-CO^{2-}).